

Question	1-3	4-5	6ab	6c	7ab	7c	8a	8b	8c	9a	9bc	9d	Total
P4	/3		/5		/6			/3		/3			/20
H5		/1				/6	/5				/7		/19
H6		/1		/7									/8
H7									/5				/5
H9												/2	/2
													/54

Section I

5 marks

Attempt Questions 1 – 5

Use the multiple-choice answer sheet for Questions 1–5.

- 1 A parabola has its focus at $(0, 2)$. The equation of its directrix is $x = -4$. The equation of the parabola is:
- (A) $x^2 = 16(y - 2)$ (B) $(y - 2)^2 = 16x$
- (C) $(y - 2)^2 = 8(x + 2)$ (D) $(x + 2)^2 = 8y$
- 2 The minimum value of $x^2 - 7x + 10$ is:
- (A) 10 (B) $-\frac{7}{2}$ (C) $-\frac{9}{4}$ (D) 0
- 3 The locus of point(s) equidistant from the points $(p, 0)$ and $(0, p)$ is:
- (A) The point $(\frac{p}{2}, \frac{p}{2})$
- (B) The line $x = y$
- (C) The line $x = -y$
- (D) The line $x = y$ for $p > 0$ and $x = -y$ for $p < 0$
- 4 The line $2x + 3y = 5$ is a tangent to the curve $y = f(x)$ at $(-2, 3)$. The same line is also a normal to the curve at $(1, 1)$. The value of $f'(1)$ could be:
- (A) $-\frac{2}{3}$ (B) $\frac{2}{3}$ (C) $-\frac{3}{2}$ (D) $\frac{3}{2}$

- 5 Consider a curve with the following properties:

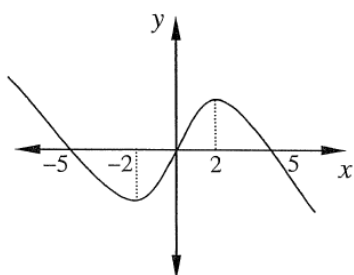
$g(x)$ is odd.

$g(5) = 0$ and $g'(2) = 0$.

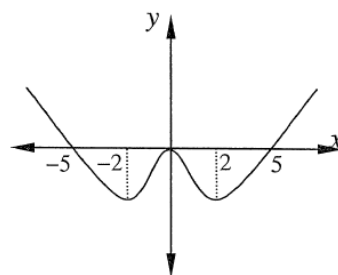
$g'(x) > 0$ for $x > 2$.

Which of the following could be the graph of $g(x)$?

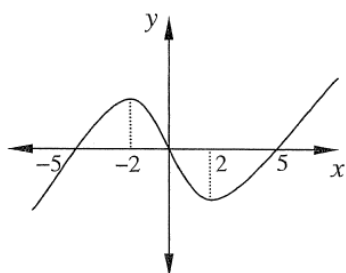
(A)



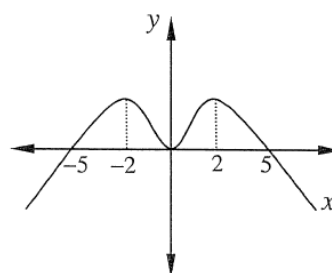
(B)



(C)



(D)



Section II

49 marks

Attempt Questions 6 – 9

Answer each question in a SEPARATE writing booklet.

In Questions 6 – 9, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (12 marks) Use a SEPARATE writing booklet.

- (a) Find the equation of the parabola with vertex at $(-3, 2)$ and focus at $(-3, 6)$. 2
- (b) A point moves such that its distance from $A(-1, 5)$ is twice its distance from $B(2, -4)$. Find the equation of its locus. 3
- (c) Consider the function $f(x) = 3x - x^3 + 5$.
- (i) Find the coordinates of the stationary points of $f(x)$ and determine their nature. 3
- (ii) Find any inflexion points of $f(x)$. 2
- (iii) Sketch the graph of $f(x)$ showing all stationary and inflexion points. You are not required to find the x -intercepts. 2

Question 7 (12 marks) Use a SEPARATE writing booklet.

- (a) Find the value(s) of K , for which the parabola $y = 6x^2$ does not meet the line $16x - y + K = 0$. 2

Question 7 (continued)

(b) α and β are the roots of the equation $2x^2 - 11x + 3 = 0$. Evaluate :

(i) $\alpha + \beta$. 1

(ii) $\alpha\beta$. 1

(iii) $(\alpha - 1)(\beta - 1)$. 2

(c) Find the primitive function of the following:

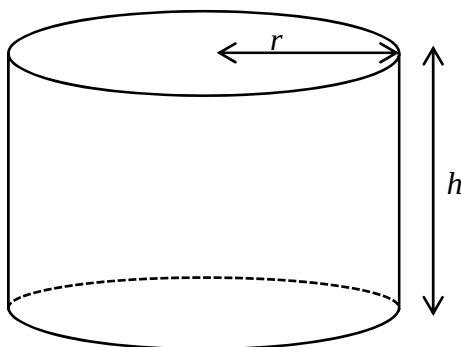
(i) $2x^3 - 3x^2 + 5$. 2

(ii) $\frac{1}{\sqrt{2x+3}}$. 2

(iii) $\frac{3x - 2x^2}{x^4}$. 2

Question 8 (13 marks) Use a SEPARATE writing booklet.

(a) A cylindrical water tank is open at the top and has a fixed volume of $V \text{ cm}^3$. The radius of the cylinder is $r \text{ cm}$ and the perpendicular height is $h \text{ cm}$.



(i) Write an expression for the height h of the cylinder in terms of V and r . 1

(ii) Show that the surface area of the tank is given by $S = \frac{2V}{r} + \pi r^2$. 1

(iii) Show that S is a minimum when $r = \left(\frac{V}{\pi}\right)^{\frac{1}{3}}$. 3

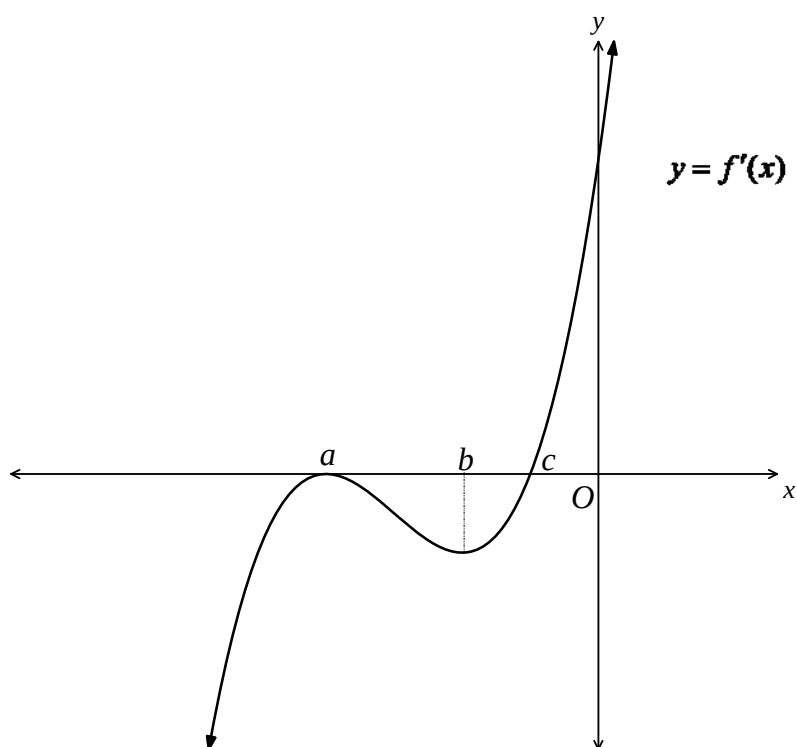
Question 8 (continued)

- (b) By using a suitable substitution, or otherwise, solve the following equation for x .

$$2(x^2 + 1)^2 - 11(x^2 + 1) + 5 = 0.$$

3

- (c) The graph of the gradient function $y = f'(x)$ is given below.



- | | | |
|-------|--|----------|
| (i) | For what values of x is the original function $f(x)$ increasing? | 1 |
| (ii) | Where is $f(x)$ concave up? | 2 |
| (iii) | Where does $f(x)$ have a turning point? | 1 |
| (iv) | What feature exists on the graph of $f(x)$ at $x = a$? | 1 |

Question 9 (12 marks) Use a SEPARATE writing booklet.

- (a) Find the values of A , B and C such that :

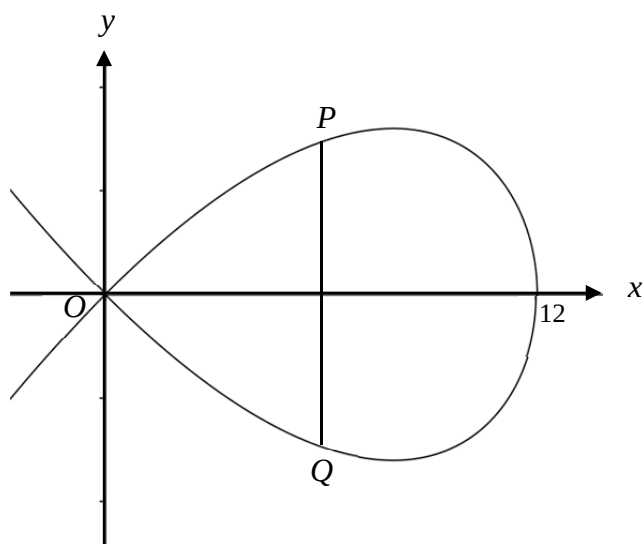
$$x^2 - 3x + 5 \equiv A(x-1) + Bx(x+1) + Cx \quad 3$$

- (b) The second derivative of a function $f''(x) = 6x - 4$ and $f(x)$ has a stationary point at $\left(\frac{2}{3}, \frac{2}{3}\right)$. Find the equation of $f(x)$. 3

- (c) PQ is a chord, parallel to the y -axis, of the loop of the curve $y^2 = x^2(12 - x)$, as shown below, where $0 < x < 12$.

- (i) Show that y^2 has a maximum in this domain at $x = 8$. 3

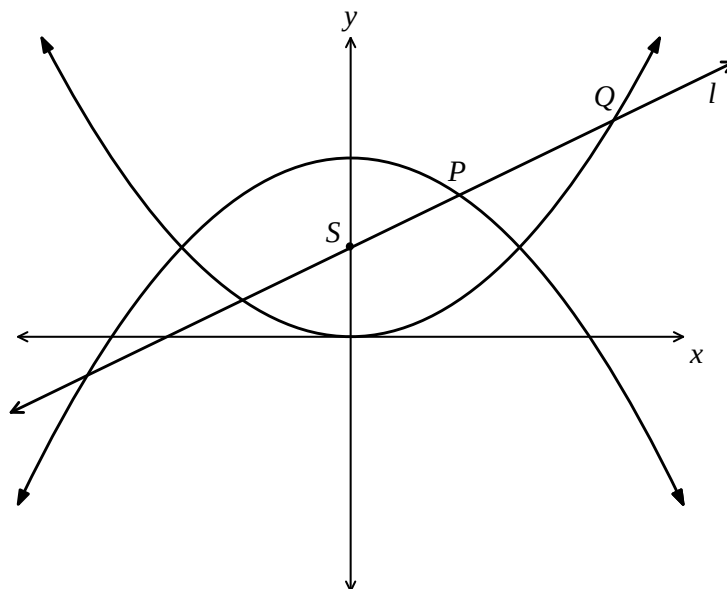
- (ii) Hence, or otherwise, find the maximum length of the chord PQ . 1



Question 9 continues on page 8

Question 9 (continued)

- (d) The diagram below, shows two parabolas $x^2 = 4y$ and $x^2 = -4(y - 2)$. They have a common focus at S .



The line l passes through S and cuts the two parabolas at $P(a,b)$ and $Q(c,d)$ as shown.

By considering the locus definition of a parabola, or otherwise, show that the length of $PQ = b + d - 2$.

2

End of paper

Section I**1 C**

Focus to the right of directrix. $\therefore$ Sideways facing parabola with focal length 2 and vertex at $(-2, 2)$.

$$(y-2)^2 = 8(x+2)$$

2 C

Minimum occurs on axis of symmetry

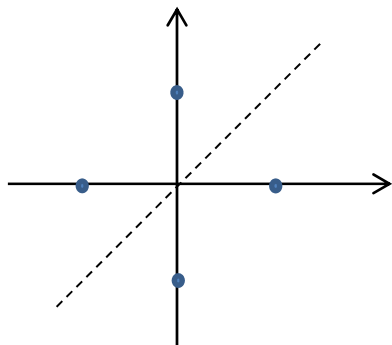
$$\text{Axis} = -\frac{b}{2a} = \frac{7}{2}$$

$$\text{Sub } x = \frac{7}{2} \quad y = -\frac{9}{4}$$

3 B

Dots show possible positions of $(p, 0)$ and $(0, p)$ for positive and negative p .

Dotted line shows locus

**4 D**

Gradient of line is $-\frac{2}{3}$ which is the gradient of the normal at $x=1$. The gradient of tangent at $x=1$ is $f'(1)$. $\therefore$ gradient of line and $f'(1)$ multiply to -1 . $\therefore f'(1) = \frac{3}{2}$

5 C

Cannot be B or D as they are even functions (symmetric about y-axis). Only C has

$$g'(x) > 0 \text{ for } x > 2$$

Section II**Question 6**

(a) Focal length = 4 and parabola faces up
 $(x+3)^2 = 16(y-2)$

(b) $PA = 2.PB \Rightarrow PA^2 = 4.PB^2$
 $(x+1)^2 + (y-5)^2 = 4[(x-2)^2 + (y+4)^2]$
 $x^2 + 2x + 1 + y^2 - 10y + 25$
 $= 4x^2 - 16x + 16 + 4y^2 + 32y + 64$
 $3x^2 - 18x + 3y^2 + 42y + 54 = 0$
 $x^2 - 6x + y^2 + 14y + 18 = 0$

is the equation of the locus of P .

(c) Stationary points occur when $f'(x) = 0$

$$3 - 3x^2 = 0$$

$$3(1 - x^2) = 0$$

$$x = \pm 1$$

$\therefore (-1, 3)$ and $(1, 7)$ are stationary points.

$$f''(x) = -6x$$

$$\text{At } (-1, 3) \quad f''(x) = -6 \times -1 = 6 > 0$$

$$\text{At } (1, 7) \quad f''(x) = -6 \times 1 = -6 < 0$$

$\therefore (-1, 3)$ is a point of local minimum and $(1, 7)$ is a point of local maximum.

(ii) Inflexion points occur when $f''(x) = 0$ and concavity changes.

$$f''(x) = 0 \Rightarrow -6x = 0 \Rightarrow x = 0$$

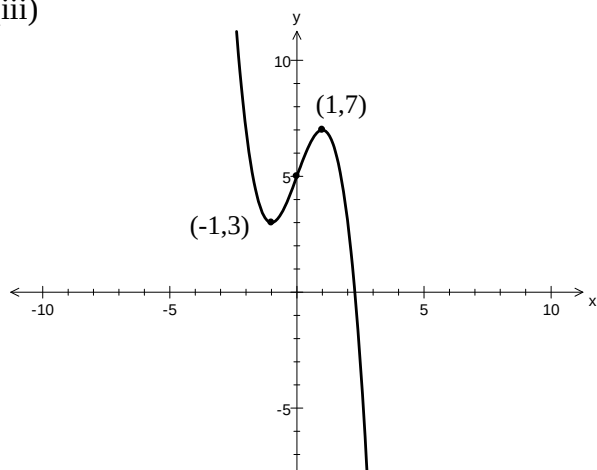
$\therefore$ there is a possible point of inflexion at $x = 0$.

Test change in concavity:

x	-0.5	0	+0.5
$f''(x)$	3	0	-3
	>0		<0
	Concave up		Concave down

There is a change in concavity. $\therefore (0, 5)$ is a point of inflexion.

(iii)

**Question 7**

- (a) For points of intersection, we solve the line and the parabola simultaneously.

$$y = 6x^2 \text{ and } y = 16x + K$$

$$6x^2 = 16x + K$$

$$6x^2 - 16x - K = 0$$

If they do not intersect, this equation has no solutions. i.e., $\Delta < 0$

$$256 - 4 \times 6 \times -K < 0$$

$$24K < -256$$

$$K < -\frac{32}{3}$$

(b) (i) $\alpha + \beta = \frac{11}{2}$

(ii) $\alpha\beta = \frac{3}{2}$

(iii) $(\alpha - 1)(\beta - 1) = \alpha\beta - \alpha - \beta + 1$

$$= \alpha\beta - (\alpha + \beta) + 1$$

$$= \frac{3}{2} - \frac{11}{2} + 1$$

$$= -3$$

(c) (i) Let $f(x) = 2x^3 - 3x^2 + 5$

$$F(x) = \frac{x^4}{2} - x^3 + 5x + C$$

(ii) Let $f(x) = \frac{1}{\sqrt{2x+3}} = (2x+3)^{-\frac{1}{2}}$

$$\begin{aligned} F(x) &= \frac{(2x+3)^{\frac{1}{2}}}{\frac{1}{2} \times 2} \\ &= \sqrt{2x+3} + C \end{aligned}$$

(iii) $f(x) = \frac{3}{x^3} - \frac{2}{x^2} = 3x^{-3} - 2x^{-2}$

$$F(x) = -\frac{3}{2x^2} + \frac{2}{x} + C$$

Question 8

(a) (i) $V = \pi r^2 h \Rightarrow h = \frac{V}{\pi r^2}$

$$\begin{aligned} \text{(ii) } S &= 2\pi rh + \pi r^2 \\ &= 2\pi r \times \frac{V}{\pi r^2} + \pi r^2 \\ &= \frac{2V}{r} + \pi r^2 \end{aligned}$$

(iii) Stationary points occur when $\frac{dS}{dr} = 0$

i.e. when $-\frac{2V}{r^2} + 2\pi r = 0$

$$2\pi r = \frac{2V}{r^2}$$

$$r^3 = \frac{2V}{2\pi} = \frac{V}{\pi}$$

$$\therefore r = \left(\frac{V}{\pi}\right)^{\frac{1}{3}}$$

$$\frac{d^2S}{dr^2} = \frac{4V}{r^3} + 2\pi > 0 \quad (\because r > 0, V > 0)$$

Or at $r = \left(\frac{V}{\pi}\right)^{\frac{1}{3}}; \frac{d^2S}{dr^2} = 4V \div \frac{V}{\pi} + 2\pi = 6\pi > 0$

$$\therefore S \text{ has a minimum at } r = \left(\frac{V}{\pi}\right)^{\frac{1}{3}}$$

(b) Let $u = x^2 + 1$

$$2u^2 - 11u + 5 = 0$$

$$(2u - 1)(u - 5) = 0$$

$$u = \frac{1}{2} \text{ or } u = 5$$

$$x^2 + 1 = \frac{1}{2} \text{ or } x^2 + 1 = 5$$

$$x^2 = -\frac{1}{2} \text{ or } x^2 = 4$$

$$x^2 = -\frac{1}{2} \Rightarrow \text{no real solns}$$

$$x^2 = 4 \Rightarrow x = \pm 2$$

(c) (i) $x > c$

(ii) $x < a$ and $x > b$

(iii) A minimum turning point at $x = c$

(iv) A stationary point of inflexion

Question 9(a) Compare coefficients of x^2 in LHS and RHS:

$$B = 1$$

$$\text{Sub } x = 0:$$

$$5 = -A \Rightarrow A = -5$$

$$\text{Sub } x = 1:$$

$$1 - 3 + 5 = 2B + C$$

$$C = 3 - 2 = 1$$

$$\therefore A = -5; B = 1; C = 1$$

(b) $f''(x) = 6x - 4$

$$f'(x) = 3x^2 - 4x + C$$

$$f'\left(\frac{2}{3}\right) = 0$$

$$3 \times \frac{4}{9} - 4 \times \frac{2}{3} + C = 0$$

$$C = \frac{4}{3}$$

$$\therefore f'(x) = 3x^2 - 4x + \frac{4}{3}$$

$$f(x) = x^3 - 2x^2 + \frac{4}{3}x + D$$

$$f\left(\frac{2}{3}\right) = \frac{2}{3}$$

$$\frac{8}{27} - 2 \times \frac{4}{9} + \frac{4}{3} \times \frac{2}{3} + D = \frac{2}{3}$$

$$D = \frac{10}{27}$$

$$f(x) = x^3 - 2x^2 + \frac{4}{3}x + \frac{10}{27}$$

(c) (i) Let $s = y^2 = x^2(12 - x)$

$$s = 12x^2 - x^3$$

Stationary points occur when $s' = 0$

$$24x - 3x^2 = 0$$

$$3x(8 - x) = 0$$

$$x = 0 \text{ or } x = 8$$

$$\text{But } 0 < x < 12$$

Only stationary point within domain is $x = 8$

$$s'' = 24 - 6x$$

$$\text{At } x = 8 \quad s'' = 24 - 6 \times 8 = -24 < 0$$

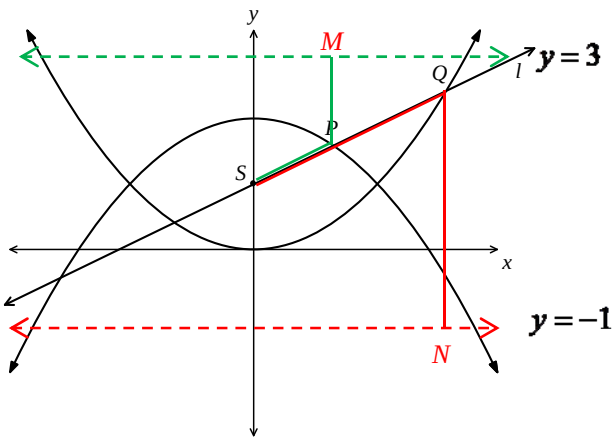
 $\therefore s$ has a maximum at $x = 8$ i.e. y^2 has a maximum at $x = 8$.

(ii) At $x = 8$, $y^2 = 8^2 \times (12 - 8) = 256$

$$y = \pm 16$$

$$\text{Length of chord PQ} = 2 \times 16 = 32 \text{ units}$$

- (d) In the diagram below, add the two directrices at $y = -1$ and $y = 3$. Draw perpendiculars PM and QN to the directrices from P and Q .



Now, by the locus definition of the parabola
 $SQ = QN = d + 1$

And $SP = PM = 3 - b$

$$\begin{aligned} PQ &= SQ - SP \\ &= (d + 1) - (3 - b) \\ &= b + d - 2 \end{aligned}$$

End of solutions